

DISEASES CAUSED BY EXFOLIATIVE TOXINS: STAPHYLOCOCCAL SCALDED- SKIN SYNDROME

Staphylococcal scalded-skin syndrome (SSSS) is a generalized exanthematous disease caused by *Staphylococcus aureus* that results in cutaneous tenderness, widespread superficial blistering, and epidermal denudation. It primarily affects infants and young children, although it can occur in adults with renal insufficiency or immunocompromised states.

Etiology and Pathogenesis

Exfoliative toxins (ETs) are serine proteases produced by certain strains of *S. aureus*, typically belonging to phage group II. The two main types are exfoliatin A (ETA) and exfoliatin B (ETB). ETA is chromosomally encoded and usually associated with localized disease, while ETB is plasmid-encoded and often linked to more generalized forms.

These toxins specifically cleave desmoglein-1, a key adhesion protein in the desmosomes of the granular layer of the epidermis. This cleavage leads to loss of cell-to-cell adhesion (acantholysis), resulting in intraepidermal splitting and the formation of flaccid bullae. The basement membrane remains intact, allowing for rapid re-epithelialization once the toxin is cleared.

The systemic spread of the toxin via the bloodstream explains the widespread cutaneous manifestations despite a localized source of infection, which may be subtle (e.g., conjunctivitis, nasopharyngitis, or a small skin focus).

Clinical Features

- **Prodrome:** Fever, irritability, and diffuse erythema.
- **Skin findings:** - Widespread erythroderma with tenderness - Positive Nikolsky sign (lateral pressure on the skin causes sloughing) - Formation of large, flaccid bullae that rupture easily - Desquamation resembling a scalded appearance, particularly in flexural areas and periorificial regions
- **Mucous membranes:** Typically spared (a key feature distinguishing SSSS from toxic epidermal necrolysis)

Diagnosis

- Clinical presentation and history (especially age group and rapid progression)
- Skin biopsy: Shows cleavage within the granular layer of the epidermis without significant inflammation or necrosis
- Cultures: May identify *S. aureus* from a primary site (e.g., nose, throat, conjunctiva), though blood cultures are typically negative
- PCR or immunoassays for exfoliative toxins (available in some reference laboratories)

Treatment

- **Antistaphylococcal antibiotics:** Initial therapy with parenteral agents such as nafcillin, oxacillin, or cefazolin; vancomycin if MRSA is suspected
- **Supportive care:** Fluid and electrolyte management, pain control, and wound care similar to burn management
- **Isolation:** To prevent spread to other vulnerable individuals

Prognosis is generally excellent with prompt treatment, as re-epithelialization occurs within 5–7 days.

GRAM-POSITIVE INFECTIONS ASSOCIATED WITH TOXIN PRODUCTION

AT A GLANCE

- *Staphylococcus aureus* and *Streptococcus pyogenes* are major Gram-positive pathogens that cause disease through toxin production.
- Toxins involved include exfoliative toxins, superantigens (e.g., toxic shock syndrome toxin-1, streptococcal pyrogenic exotoxins), and pore-forming toxins (e.g., alpha-toxin, streptolysin O).
- Host factors such as skin/mucosal integrity, immune competence, and presence of foreign bodies significantly influence susceptibility.
- Superantigens cause nonspecific activation of T cells, leading to massive cytokine release and systemic illness (e.g., toxic shock syndrome).
- Early recognition and targeted antimicrobial therapy, along with supportive care, are critical for favorable outcomes.